SMQTT (Secure MQTT)

SMQTT is an enhanced version of MQTT that prioritizes security for IoT applications, particularly for resource-constrained devices. It integrates advanced cryptographic mechanisms, such as attribute-based encryption (ABE), to ensure secure, fine-grained access control.

Features

1. Attribute-Based Encryption (ABE): Encrypts messages based on attributes (e.g., device role, location), allowing only authorized subscribers to decrypt.
2. Lightweight Cryptography: Uses efficient algorithms (e.g., elliptic curve cryptography, ECC) to minimize computational overhead on low-power devices (e.g., 8-bit MCUs).
3. End-to-End Security: Messages remain encrypted through the broker, ensuring confidentiality.
4. Policy-Based Access: Defines access policies (e.g., “only managers in building A”) enforced by ABE.
5. Compatibility: Retains MQTT’s publish-subscribe model, integrating with existing brokers.
6. Scalability: Supports secure communication in large IoT networks with thousands of devices.
7. Low Overhead: Adds minimal latency compared to standard MQTT, suitable for constrained networks.

Working Principle

- SMQTT builds on MQTT by adding a security layer:

1. Key Distribution: A trusted key management authority generates public/private key pairs based on attributes (e.g., role=admin, location=room1).
2. Message Encryption: The publisher encrypts the message with an ABE policy (e.g., “accessible by devices with role=admin”).
3. Broker Routing: The broker forwards the encrypted message to subscribers without decrypting it, preserving MQTT’s topic-based routing.
4. Decryption: Subscribers with matching a tributes (e.g., role=admin) use their private keys to decrypt the message.
5. Authentication: Optional client authentication (e.g., via certificates) ensures only authorized devices connect.

- Example:
  - A sensor in a smart factory publishes encrypted data to secure /factory/temperature with a policy “accessible by maintenance team in sector B.”
  - Only devices with attributes role=maintenance and sector=B can decrypt the message, even if others subscribe to the topic.

Applications

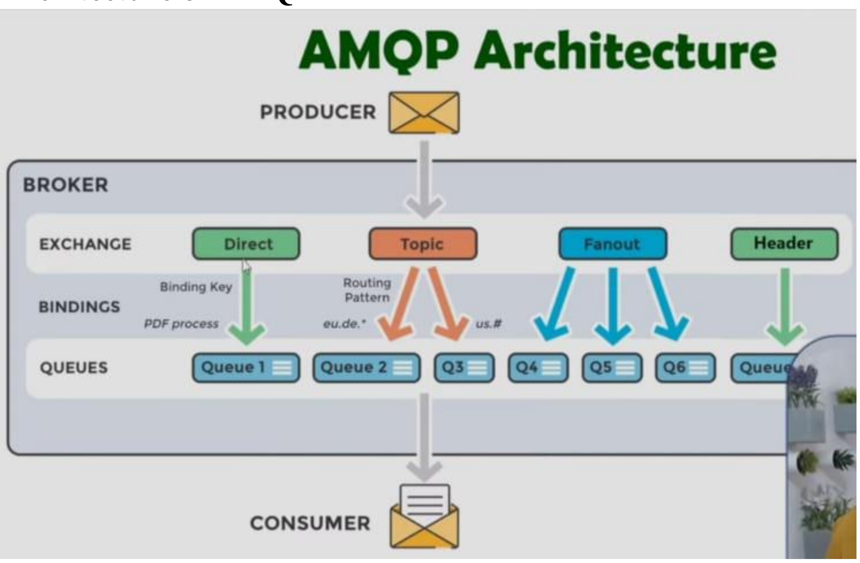
1. Smart Grids: Securely transmit energy consumption data to authorized control systems.
2. Healthcare: Protect sensitive patient data (e.g., heart rate) from unauthorized access.
3. Industrial IoT: Secure machine-to-machine communication in factories with role based access.
4. Smart Homes: Restrict access to critical controls (e.g., smart locks) to family members.
5. Defence IoT: Secure sensor data in military applications with strict access policies.

AMQP (Advanced Message Queuing Protocol)
AMQP is a robust, open-standard protocol for message-oriented middleware, designed for reliable, asynchronous message queuing in enterprise and IoT applications. It supports complex routing and guaranteed delivery, making it suitable for high-throughput systems.

Basics of AMOP

- AMQP-Advanced Message Queuing Protocol
- It is an open standard application layer protocol for message-oriented middleware.
- AMQP is mainly used in transaction messages between servers and users.
- Mostly it is used in banking servers and banking applications.
- It uses TCP for reliable communication.
- End points acknowledge acceptance of each message.
- It follows Public Subscriber model for communication.

Architecture of AMQP



The image shows the **AMQP Architecture** (Advanced Message Queuing Protocol), which is a standard protocol used in **message-oriented middleware**. It enables systems to communicate by sending messages through a **broker** system, decoupling the sender (producer) and receiver (consumer).

1. Producer

- The **producer** is the sender of messages. It sends messages to the **broker** (not directly to the consumer).
- The producer does **not need to know** who will consume the message.

2. Broker

- The **broker** is the middleware that routes messages from producers to consumers using **exchanges**, **bindings**, and **queues**.

3. **Exchange**

- An **exchange** is responsible for routing messages to the correct **queue(s)** based on specific rules. There are different types:
  - **Direct**: Routes messages with a specific binding key to the matching queue.
  - **Topic**: Routes messages based on a **pattern-matching** routing key.

- **Fanout**: Broadcasts the message to **all bound queues**, ignoring routing keys.
- **Header**: Routes messages based on **message headers**, not routing keys.

4. Bindings

- A **binding** is the link between an exchange and a queue.
- It uses **routing logic**, such as:
  - Binding Key: e.g., `"PDF process"` for direct exchange.
  - Routing Pattern: e.g., `"eu.de.*"` or `"us.#"` for topic exchange.

5. **Queues**

- **Queues** hold messages until they are consumed.
- Each queue can be linked to one or more exchanges with specific binding rules.
- In the diagram:
  - **Queue 1** gets messages from the **Direct** exchange.
  - **Queue 2**, **Q3** get messages from the **Topic** exchange based on different routing patterns.
  - **Q4**, **Q5**, **Q6** get messages from **Fanout** (all of them get the same message).
  - **Queue 7** is connected to **Header** exchange (based on header content).

3. Consumer

- The **consumer** is the receiver of messages.
- It reads messages from the queue.
- Like the producer, it doesn't need to know who sent the message.

Features

1. Reliable Delivery: Ensures messages are delivered with acknowledgments and transactions.
2. Flexible Rou ng: Supports multiple exchange types (direct, topic, fanout, headers) for precise message distribution.
3. Asynchronous Messaging: Decouples producers and consumers, enabling high scalability.
4. Security: Uses TLS for encryption, SASL for authentication, and access control (e.g., virtual hosts in RabbitMQ).
5. Message Durability: Persists messages in queues un l consumed, surviving broker restarts.
6. Cross-Platform: Interoperable across languages (e.g., Python, Java) and brokers (e.g., RabbitMQ, Apache Qpid).
7. High Throughput: Handles thousands of messages per second in distributed systems.
8. Transactional Support: Groups messages into atomic transactions for consistency.
9. Message Prioritization: Allows high-priority messages (e.g., alerts) to be processed first.

Components

1. Producer: Sends messages to an exchange (e.g., IoT sensor publishing temperature).
2. Consumer: Retrieves messages from a queue (e.g., analytics server processing data).
3. Exchange: Routes messages to queues based on routing rules:
   - Direct: Matches exact routing key (e.g., sensor. temperature).

- o Topic: Matches pa erns (e.g., sensor.*.temp for all rooms).
- o Fanout: Broadcasts to all bound queues (e.g., alerts to multiple systems).
- o Headers: Routes based on message header attributes (e.g., priority=high).

4. Queue: Stores messages until consumed, with options for durability and expiration.
5. Broker: Manages exchanges, queues, and bindings (e.g., RabbitMQ, ActiveMQ).
6. Binding: Links exchanges to queues with routing keys or conditions.
7. Virtual Host: Logical isolation within a broker for mul -tenant setups.

AMQP Frames

- Frame header: The header has a fixed size (8 bytes), and it is mandatory, as it contains the information needed to parse the rest of the frame itself-for example the total frame size and the frame type.
- Extended header: A variable header that depends on the frame type.
- Frame body: A sequence of bytes that has a format that depends on the frame type.
- Frame types can be Open, Close, begin, end, attach, detach, transfer, flow and disposition.

Communication

- Model: Producer-exchange-queue-consumer, with asynchronous message queuing:

1. Connection: Producer/consumer connects to the broker, sending a protocol header and negotiating parameters (e.g., heartbeat interval).
2. Channel Setup: Opens a logical channel for communication (method frame: channel. Open).
3. Exchange/Queue Setup: Declares exchanges and queues (e.g., exchange. declare, queue. declare) and binds them (e.g., queue. bind with routing key sensors.*).
4. Publishing: Producer sends a message via method frame (basic. publish), content header (properties), and body frame(s) to an exchange.
5. Rou ng: The exchange routes the message to queues based on bindings (e.g., topic exchange matches sensors.room1.temp to sensors.*.temp).
6. Consumption: Consumer retrieves messages from a queue (e.g., basic. consume), processes them, and sends acknowledgments (basic ack) for reliability.

7. Durability: Persistent queues/messages survive broker restarts; transactions (tx. select, tx. commit) ensure atomicity.

- Example:
  - A sensor (producer) publishes a message to a topic exchange with routing key sensors.room1.temp and payload {"temp": 25.5}.
  - The broker routes it to two queues bound to sensors.*.temp (e.g., for analytics and alerts).
  - An analytics app (consumer) retrieves the message, processes it, and sends an ACK to the broker.

Applications

1. Industrial IoT: Queues sensor data for real- me analytics or maintenance alerts.
2. Smart Cities: Routes traffic or environmental data to control systems.
3. Logistics: Tracks shipments with reliable, high-throughput messaging.
4. Financial Systems: Processes stock trades or payment transactions with guaranteed delivery.
5. Smart Grids: Coordinates energy devices with asynchronous, durable messaging.
6. Micro services: Integrates IoT with enterprise systems via message queues.

Introduction of IEEE 802.15.4 Technology

IEEE 802.15.4 is a low-cost, low-data-rate wireless access technology for devices that are operated or work on batteries. This describes how low-rate wireless personal area networks (LR-WPANs) function.

IEEE 802.15.4e:

802.15.4e for industrial applications and 802.15.4g for the smart utility networks (SUN)
The 802.15.4e improves the old standard by introducing mechanisms such as time slotted access, multichannel communication and channel hopping.

- **IEEE 802.15.4e introduces the following general functional enhancements:**
  1. Low Energy (LE): This mechanism is intended for applications that can trade latency for energy efficiency. It allows a node to operate with a very low duty cycle.
  2. Information Elements (IE) It is an extensible mechanism to exchange information at the MAC sublayer.
  3. Enhanced Beacons (EB): Enhanced Beacons are an extension of the 802.15.4 beacon frames and provide a greater flexibility. They allow to create application-specific frames.
  4. Multipurpose Frame: This mechanism provides a flexible frame format that can address a number of MAC operations. It is based on IEs.
  5. MAC Performance Metric: It is a mechanism to provide appropriate feedback on the channel quality to the networking and upper layers, so that appropriate decision can be taken.
  6. Fast Association (FastA) The 802.15.4 association procedure introduces a significant delay in order to save energy. For time-critical application latency has priority over energy efficiency.

- IEEE 802.15.4e defines five new MAC behavior modes.

1. Time Slotted Channel Hopping (TSCH): It targets application domains such as industrial automation and process control, providing support for multi-hop and multichannel communications, through a TDMA approach.

2. Deterministic and Synchronous Multi-channel Extension (DSME): It is aimed to support both industrial and commercial applications.

3. Low Latency Deterministic Network (LLDN): Designed for single-hop and single channel networks

4. Radio Frequency Identification Blink (BLINK): It is intended for application domains such as item/people identification, location and tracking.

5. Asynchronous multi-channel adaptation (AMCA): It is targeted to application domains where large deployments are required, such as smart utility networks, infrastructure monitoring networks, and process control networks.

Properties:

1. Standardization and alliances: It specifies low-data-rate PHY and MAC layer requirements for wireless personal area networks (WPAN).
IEEE 802.15. Protocol Stacks include:

- **ZigBee:** ZigBee is a Personal Area Network task group with a low rate task group 4. It is a technology of home networking. ZigBee is a technological standard created for controlling and sensing the network. As we know that ZigBee is the Personal Area network of task group 4 so it is based on IEEE 802.15.4 and is created by Zigbee Alliance.
- **6LoWPAN:** The 6LoWPAN system is used for a variety of applications including wireless sensor networks. This form of wireless sensor network sends data as packets and uses IPv6 - providing the basis for the name - IPv6 over Low power Wireless Personal Area Networks.
- **ZigBee IP:** Zigbee is a standards-based wireless technology that was developed for low-cost and low-power wireless machine-to-machine (M2M) and internet of things (IoT) networks.
- **ISA100.11a:** It is a mesh network that provides secure wireless communication to process control.
- **Wireless HART:** It is also a wireless sensor network technology, that makes use of time-synchronized and self-organizing architecture.
- **Thread:** Thread is an IPv6-based networking protocol for low-power Internet of Things devices in IEEE 802.15. 4-2006 wireless mesh network. Thread is independent.

2. Physical Layer: This standard enables a wide range of PHY options in ISM bands, ranging from 2.4 GHz to sub-GHz frequencies. IEEE 802.15.4 enables data transmission speeds of 20 kilobits per second, 40 kilobits per second, 100 kilobits per second, and 250 kilobits per second. The fundamental structure assumes a 10-meter range and a data rate of 250 kilobits per second. To further reduce power usage, even lower data rates are possible. IEEE 802.15.4 regulates the RF transceiver and channel selection, and even some energy and signal management features, at the physical layer. Based on the frequency range and data performance needed, there are now six PHYs specified. Four of them employ frequency hopping techniques known as Direct Sequence Spread Spectrum (DSSS). Both PHY data service and management service share a single packet structure so that they can maintain a common simple interface with MAC.

3. MAC layer: The MAC layer provides links to the PHY channel by determining that devices in the same region will share the assigned frequencies. The scheduling and routing of data packets are also managed at this layer. The 802.15.4 MAC layer is responsible for a number of functions like:

- Beaconing for devices that operate as controllers in a network.
- used to associate and dissociate PANs with the help of devices.
- The safety of the device.
- Consistent communication between two MAC devices that are in a peer-to-peer relationship.

Several established frame types are used by the MAC layer to accomplish these functions. In 802.15.4, there are four different types of MAC frames:

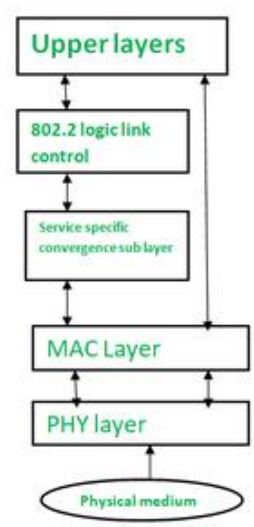
- frame of data
- Frame for a beacon
- Frame of acknowledgement
- Frame for MAC commands

4. Topology: Networks based on IEEE 802.15.4 can be developed in a star, peer-to-peer, or mesh topology. Mesh networks connect a large number of nodes. This enables nodes that would otherwise be out of range to interact with each other to use intermediate nodes to relay data.

5. Security: For data security, the IEEE 802.15.4 standard employs the Advanced Encryption Standard (AES) with a 128-bit key length as the basic encryption technique. Activating such security measures for 802.15.4 significantly alters the frame format and uses a few of the payloads. The very first phase in activating AES encryption is to use the Security Enabled field in the Frame Control part of the 802.15.4 header. For safety, this field is a single bit which is assigned to 1. When this bit is set, by taking certain bytes from its Payload field, a field known as the Auxiliary Security Header is formed following the Source Address field.

6. Competitive Technologies: The IEEE 802.15.4 PHY and MAC layers serve as a basis for a variety of networking profiles that operate in different IoT access scenarios. DASH7 is a competing radio technology with distinct PHY and MAC layers.

The architecture of LR-WPAN Device:



This diagram represents the **IEEE 802.11 (Wi-Fi) protocol stack** architecture, showing the layered structure and how data flows through different layers in wireless communication.

1. Upper Layers

- These layers represent the higher-level network protocols and applications, such as IP, TCP, HTTP, etc.
- They generate data to be transmitted or process data received from lower layers.

2. 802.2 Logic Link Control (LLC)

- This layer is part of the Data Link Layer (Layer 2 in OSI model).

- It provides **logical link control**, managing communication between the upper layers and lower sublayers.
- It handles framing, flow control, and error checking.

3. Service Specific Convergence Sub Layer (SSCS)

- This sublayer adapts services to meet specific requirements of various network types (like Ethernet, Token Ring, or wireless).
- It acts as an interface between the LLC and the MAC layer, adapting the protocol to different physical networks.

4. MAC Layer (Medium Access Control)

- Responsible for **controlling how devices access the physical medium** (e.g., wireless radio waves).
- Manages addressing, channel access, collision avoidance, and frame formatting.
- In wireless networks, it handles important functions like retransmissions and acknowledgments.

5. PHY Layer (Physical Layer)

- This is the lowest layer in this stack.
- Responsible for **transmitting raw bits over the physical medium** (e.g., radio waves, cables).
- It defines electrical, mechanical, and procedural specifications for the physical connection.

6. Physical Medium

- The actual medium through which data is transmitted (e.g., air for Wi-Fi, fiber optics, copper cables).

Near Field Communication (NFC)

Near Field Communication (NFC) is a short-range wireless communication technology that has revolutionized how devices interact in various industries, from payments to data sharing.
This NFC tutorial guide covers the protocol stack, working principles, frame structure, and different communication modes.

What is NFC (Near Field Communication)?

NFC stands for Near Field Communication. It’s a short-range, low data rate wireless communication technology operating at 13.56 MHz frequency. It’s a contactless mode of communication using electromagnetic waves.
This technology allows two devices housing an NFC chip to communicate for various applications, for example:

- Data communication between smartphones
- Verification of an authenticated person having an NFC ID card at the office by an NFC reader in the door.
- Banking payments using NFC compliant credit cards to enhance security
- Ticket booking at airports
- Automatic running of scheduled features in an NFC phone.

The following table summarizes features of wireless NFC (Near Field Communication) technology.

How NFC Works

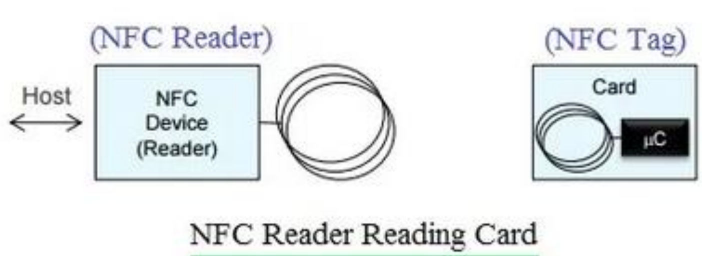


Fig:1 NFC Network

As mentioned, an NFC network consists of two devices known as the initiator device and the target device. An NFC tag can be an active or passive device. An NFC reader is always an active device. These devices operate in either active-active or active-passive modes.
In active-active mode, both NFC devices have their own power. In active-passive mode, the passive device derives its power from the received EM waves of the active device.
The basic mode of communication in NFC is half-duplex, where one NFC device transmits while the other receives. This is also referred to as "Listen before Talk."
One of the two NFC devices will function as an initiator, which first listens on the channel and transmits only when no other signal is present. The Initiator polls other devices that come closer, and the other NFC device, referred to as the target, listens and responds to the initiator as per the requested message.

NFC Network Modes: Card Emulation, Reader/Writer, Point to Point (Peer)

There are three modes in which NFC tags and readers work: Card Emulation, Reader/Writer, and Peer to Peer (or Point to Point), as explained below.

NFC Card Emulation Mode

In this mode, usually an active device reads a passive device.

EXAMPLE #1: ID cards, NFC tags, or NFC-compliant tickets are read by NFC-compliant active readers at offices or railway stations.

EXAMPLE #2: A smartphone acts as a
Smart card to allow booking tickets or performing online banking transactions or payments using a credit card reader.

Following are the steps involved in card emulation mode:

- Step 1: The NFC reader oscillates at a 13.56 MHz RF field. When the card comes near the RF field, it gets power due to EM (Electro-Magnetic) coupling and connects with the reader.
- Step 2: The reader sends commands using the RF field.
- Step 3: The card responds to the reader as requested.

NFC Reader/Writer Mode

In this mode, one device will be in reading mode, and the other will be in writing mode. They can be either active or passive devices.

- Example #1: Product Authentication
- Example #2: Smart advertising
- Example #3: Device pairing

Following are the steps involved in reader/writer mode:

- Step 1: In read mode, the NFC reader reads NFC tags, either passive or active.
- Step 2: In write mode, one NFC device writes to the other NFC device.

NFC Peer to Peer (Point to Point)

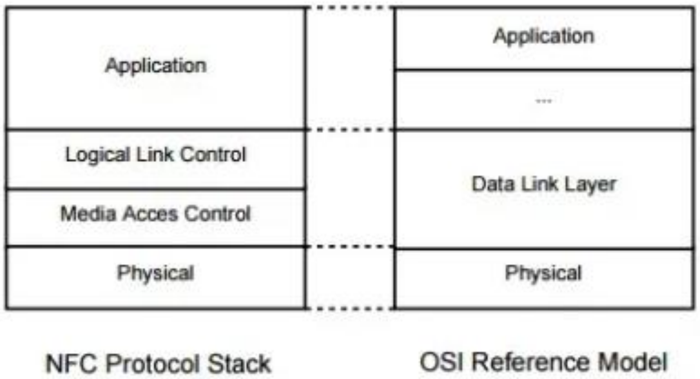
In this mode, data is transmitted from one NFC device to another in ad-hoc or peer-to-peer mode. Any device can become the initiator, and the rest will act as targets to complete the connection establishment.

- Example #1: Smartphone to smartphone communication
- Example #2: Automotive (in-car operation)
- Example #3: Social media applications or games

Following are the steps involved in peer-to-peer mode (active mode):

- Step 1: Both NFC smartphones establish pairing for data communication. Both have power and their own RF fields.
- Step 2: The initiator sends commands by modulating the RF field and switches off the RF field.
- Step 3: The target device responds to the initiator using its own RF field modulation.

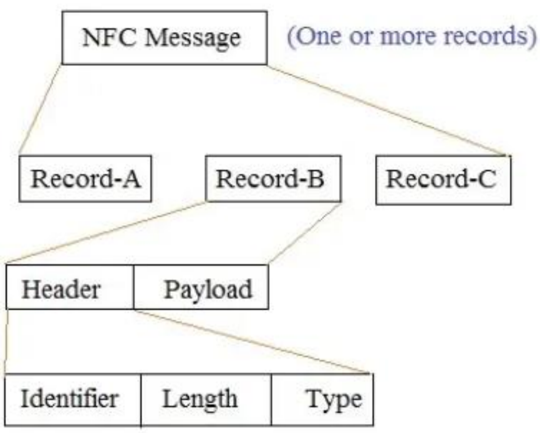
NFC Protocol Stack



The figure depicts the protocol stack of NFC in peer-to-peer mode. As shown, it consists of the application layer, data link layers (logical link control and media access control), and the physical layer.

- **Application Layer:** This layer takes care of the format of the data to be exchanged between NFC devices or between the NFC device and tags. NFC uses NDEF (NFC Data Exchange Format).
- **Data Link Layer:** This layer takes care of different modes of operation and anti-collision mechanisms. LLCP establishes communication between two peer devices.
- **Physical Layer:** This layer takes care of modulation, coding, and RF-related parameters such as frequency and power.

NFC Frame Structure



The NFC frame uses a special encapsulation format referred to as NDEF (NFC Data Exchange Format). Each NFC device transmission is known as a message. One message is composed of one or more records. Each record consists of a header part and a payload part. The header part is made of identifier, length, and type fields. The payload carries data or a URL.

Basics of COAP

- CoAP-Constrained Application Protocol.
- CoAP is an IoT Protocol.
- CoAP is designed to allow simple devices to join IoT through the low bandwidth-restricted network.
- CoAP is designed for M2M and IoT applications.
- CoAP is an Application Layer Protocol that follows the request-response model.
- CoAP runs over UDP Protocol.
- CoAP uses fewer resources than HTTP.
- COAP client can use GET, PUT, and DELETE commands during.

COAP Layers

CoAP is divided into two layers.

- Upper Layer (Application Layer) is designed for the communication method based on the Request-Response model.
- Lower Layer (UDP) is designed to deal with UDP and asynchronous messages.

Message Types COAP

CoAP Supports Four different message types:

1. **Confirmable Message (Reliable Message) (Con):**

- A Confirmable message requires a response, either a Positive Acknowledgement or a Negative Acknowledgement.
- If acknowledgement is not received then retransmissions are made until all attempts are exhausted.
- The Acknowledgement message contains the same ID as the confirmation message.
- If the server has trouble managing the incoming request, it message (RST) instead of an acknowledge message (ACK).

2. **Non-confirmable Message {Unreliable Message} (Non):**

- A Non-confirmable message doesn't require a response in the form of Acknowledgement.

- These messages do not contain critical information, like a request for a sensor measurement made in periodic basis.
- Even if one value is missed, there is not too much impact.
- Even if these messages are unreliable, they have a unique ID.

3. **Acknowledgement (Ack):**

- It is sent to acknowledge a confirmable message (Con).
- Confirms receipt of a CON message, carrying response data or error codes (e.g., 2.05 Content, 4.04 Not Found).
- Matches the CON's message ID to ensure correct pairing.
- Example: A server responds with an ACK containing 500 lux to a CON GET /sensors/light.

4. **Reset (RST):**

- It represents negative Acknowledgment.
- It generally indicates some kind of failure, like unable to process received data.
- Sent when a message is received but cannot be processed (e.g., unknown resource, malformed request).
- Indicates rejection or error without further ac on.
- Example: A client receives an RST if it sends a GET to a non-existent resource /sensors/invalid.

Message Structure:

- Header: 4 bytes (Version, Type, Token Length, Code, Message ID).
- Token: Correlates requests and responses (0–8 bytes).
- Options: Metadata (e.g., Uri-Path: sensors/temp, Content-Format: text/plain).
- Payload: Data (e.g., 25.5°C).

Process:

1. Request: Client sends a request with a method (e.g., GET /sensors/temp), message type (CON or NON), and token.
2. Response: Server replies with a response code (e.g., 2.05 Content for success, 4.04 Not Found for errors) and payload.
3. Methods:
   - GET: Retrieves resource state (e.g., current temperature).
   - POST: Creates or updates a resource (e.g., sets a new threshold).
   - PUT: Updates a resource (e.g., changes device mode).
   - DELETE: Removes a resource (e.g., deletes a log entry).
4. Observe: Clients send a GET with an Observe op on (value 0) to subscribe to resource updates; the server no files changes (e.g., temperature rises above 30°C).
5. Block-Wise Transfer: Large data (e.g., firmware updates) is split into blocks, with options like Block1 (request) and Block2 (response).

- Example:
  - A CoAP client sends a CON GET /sensors/humidity with token 0xAB12.

- The server responds with an ACK, code 2.05 Content, and payload 60% RH, using the same token.
- If the client observes /sensors/humidity, the server sends NON messages whenever humidity changes significantly.

Applications

1. Smart Homes: Control devices (e.g., lights) in Thread or 6LoWPAN networks.
2. Smart Cities: Monitor parking spaces or air quality with low-power sensors.
3. Industrial IoT: Real monitoring of machine parameters over constrained networks.
4. Wearables: Low-power data exchange for health sensors (e.g., pulse monitors).
5. Environmental Monitoring: Collect data from remote sensors (e.g., river water levels).
6. Building Automation: Manage HVAC or lighting with multicast commands.

XMPP (Extensible Messaging and Presence Protocol)

XMPP is an open, XML-based protocol for real- me messaging, presence, and request response communication. Initially developed for instant messaging, it is adapted for IoT through extensions (XEPs) for device communication and control.

Basics of XMPP

- XMPP-eXtensible Messaging and Presence Protocol
- It is a protocol (standard) that is used to build chat systems.
- It uses XML to exchange data between client and server.
- eXtensible: The protocol can be (has been) extended with new features.
- Messaging: Send one-to-one and group messages.
- Presence: See your contact's online status.

XMPP Stanzas

Stanza: It is the basic unit of communication in XMPP.
There are three types of stanzas in XMPP

1. Message: used when you want to send messages
2. Presence: used to send online status information and to control subscription status between contacts.
3. IQ (Info/Query): used to [get] some information from the server or to [set] or apply some settings.

XMPP Features

1. Real-Time Communication: Delivers messages and presence updates instantly (e.g., device status changes).
2. Extensibility: XML stanzas support custom extensions via XMPP Extension Protocols (XEPs) for IoT (e.g., XEP-0323 for sensor data, XEP-0325 for control).
3. Decentralized Architecture: Federated servers (no central authority) enhance reliability and scalability.
4. Security: Supports TLS for encryption, SASL for authentication, and end-to-end encryption (e.g., OMEMO).
5. Peer-to-Peer Sessions: XMPP supports machine-to-machine or peer-to-peer across diverse set of networks.
6. Presence Management: Tracks device availability (e.g., online, offline, busy).
7. Bidirectional Streaming: Persistent TCP connections allow continuous client-server communication.

8. Mul -Device Support: A single JID (e.g., device@domain.com) can represent multiple devices or resources.
9. High Overhead: XML-based messages are verbose, less suitable for ultra-constrained devices. Components and Communication.

Multi-User chat: XMPP supports group and conference chat.

1. Encryption: Point-to-point encryption is done by TLS {Transport Layer Security} and OTR {Off The Record messaging} is an extension of XMPP that enables encryption of messages and data.
2. Connection to other protocols: using XMPP Gateways other protocols like SMS and SMTP can also be connected with XMPP servers for different services.

Components:

- Client: Device or application identified by a JID (Jabber ID, e.g., sensor1@domain.com/resource).
- Server: Routes stanzas and manages presence (e.g., Openfire, ejabberd, Prosody).
- Stanzas: XML data units for communication:
- Message: Carries data (e.g., <message><body>{"temp": 25}</body></message>).
- Presence: Indicates status (e.g., <presence><status>online</status></presence>).
- IQ (Info/Query): Request-response for queries or commands (e.g., <iq type="get"> to retrieve device config).
- Domain: Logical grouping of clients (e.g., domain.com).

- **Communication:**

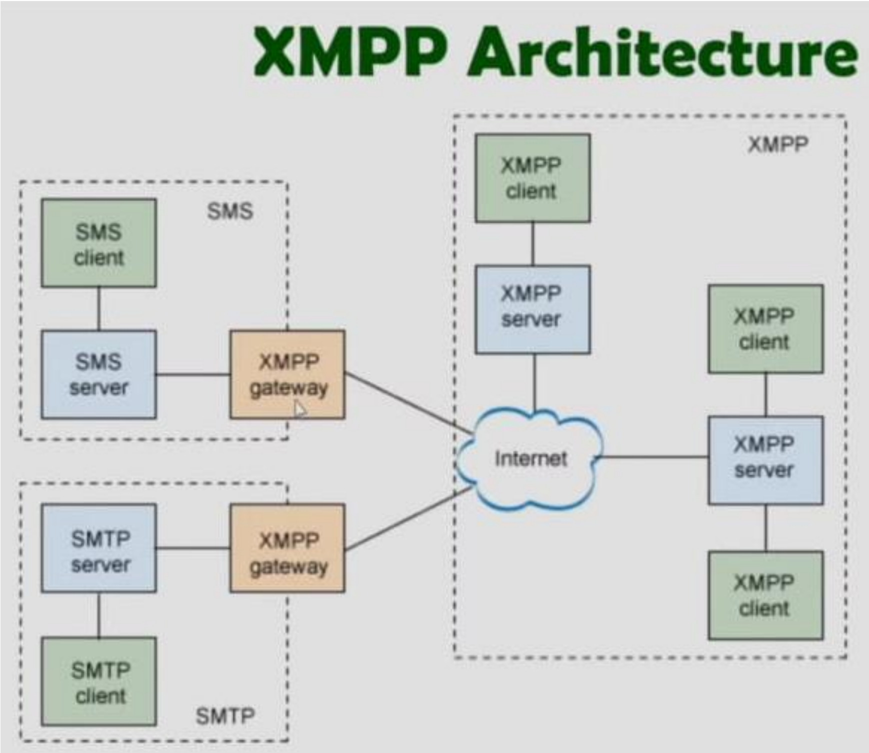
1. Connection: Client establishes a TCP connection to a server, authenticates via SASL (e.g., with a password or certificate).
2. Stanza Exchange: Clients send stanzas to other JIDs or servers:
   - Message: Sends data (e.g., sensor readings to an app).
   - Presence: Broadcasts status (e.g., device online after boot).
   - IQ: Queries or sets data (e.g., GET for current sensor value, SET for actuator control).
3. Routing: Servers forward stanzas based on JIDs, supporting federation across domains.
4. Extensions: IoT-specific XEPs enable sensor data (XEP-0323), control (XEP-0325), or provisioning (XEP-0324).

Example:

- A smart thermostat (thermostat@home.com) sends a <message> stanza with {"temp": 22.5} to a mobile app (user@home.com).
- The app responds with an IQ stanza to set the thermostat to 24°C.
- The server broadcasts a <presence> stanza when the thermostat goes offline.

Applications

1. Smart Homes: Real- me control of lights, locks, or thermostats with presence updates.
2. Industrial IoT: Machine status notifications to operators via XMPP servers.
3. Healthcare: Patient monitoring with alerts for abnormal vital signs.
4. Chat-Based IoT: Integrates devices with messaging platforms (e.g., control a lamp via a Chatbot).
5. Smart Grids: Coordinates energy devices with real- me messaging and presence.
6. Collaborative IoT: Mul -user device control in shared environments (e.g., office automation).



The image provided is a diagram of the XMPP Architecture (Extensible Messaging and Presence Protocol), showing how XMPP integrates with other communication systems like SMS and SMTP (email).

1. XMPP Clients and Servers (Right side)
   - XMPP Clients: These are applications or users connected to the XMPP network (e.g., messaging apps like Jabber, Pidgin, etc.).
   - XMPP Servers: These manage communication between XMPP clients. They handle message delivery, presence info, and routing.
   - Communication: Clients connect to servers. Servers communicate with each other via the Internet to allow messages to pass across domains.
2. Gateways (Center)
   - XMPP Gateways act as bridges between XMPP and other communication systems (e.g., SMS, Email/SMTP).
   - They translate the message formats and protocols so users in the XMPP system can communicate with users on other systems (like sending an XMPP message as an SMS or email).
3. SMS System (Left top)
   - SMS Client: A user or device sending/receiving text messages.
   - SMS Server: Handles SMS message routing.
   - XMPP Gateway: Connects the SMS server to the XMPP network, allowing SMS users to interact with XMPP users.
4. SMTP System (Left bottom)
   - SMTP Client: An email-sending application (e.g., a mail client).
   - SMTP Server: Sends and receives emails using SMTP protocol.
   - XMPP Gateway: Enables communication between email systems and the XMPP network (e.g., XMPP user can send a message that is received as an email).

Internet Connectivity

All systems (XMPP servers, SMS/XMPP gateways, SMTP/XMPP gateways) are connected via the Internet, ensuring interoperability and communication across platforms.

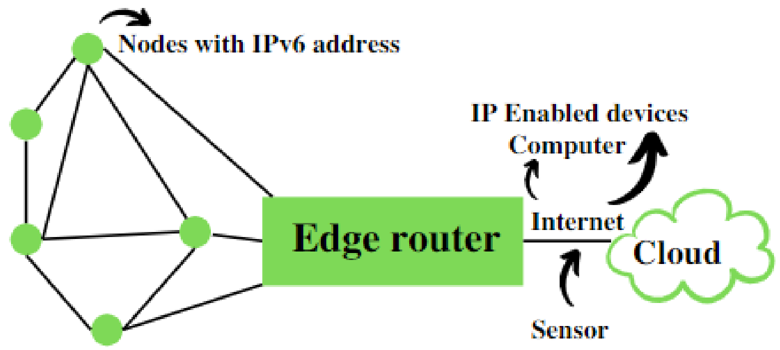
What is 6LoWPAN?

6LoWPAN stands for IPv6 over Low-Power Wireless Personal Area Networks. It's a communication protocol designed to enable small, low-power devices to connect over wireless networks. In this article, we will discover everything about 6LoWPAN and its various advantages and disadvantages.

What is 6LoWPAN?

6LoWPAN is an IPv6 protocol, and It's extended from IPv6 over Low Power Personal Area Network. As the name itself explains the meaning of this protocol is that this protocol works on Wireless Personal Area Network. WPAN is a Personal Area Network (PAN) where the interconnected devices are centered around a person's workspace and connected through a wireless medium. You can read more about WPAN at WPAN. 6LoWPAN allows communication using the IPv6 protocol. IPv6 is Internet Protocol Version 6 is a network layer protocol that allows communication to take place over the network. It is faster and more reliable and provides a large number of addresses.

6LoWPAN initially came into existence to overcome the conventional methodologies that were adapted to transmit information. But still, it is not so efficient as it only allows for the smaller devices with minimal processing ability to establish communication using one of the Internet Protocols, i.e., IPv6. It has very low cost, short-range, low memory usage, and low bit rate. It comprises an Edge Router and Sensor Nodes. Even the smallest of the IoT devices can now be part of the network, and the information can be transmitted to the outside world as well. For example, LED Streetlights.



It is a technology that makes the individual nodes IP-enabled. 6LoWPAN can interact with 802.15.4 devices and also other types of devices on an IP Network. For example, Wi-Fi. It uses AES 128 link layer security, which AES is a block cipher having key size of 128/192/256 bits and encrypts data in blocks of 128 bits each. This is defined in IEEE 802.15.4 and provides link authentication and encryption.

Architecture of 6LOWPAN:

- 6LOWPAN Technology makes individual nodes IP-enabled.
- Edge Router is the core of the 6LOWPAN network which connects the 6LOWPAN network with the internet.
- 6LOWPAN can communicate with ZigBee devices (IEEE 802.15.4) and Wi-Fi devices (IEEE 802.11).
- 6LOWPAN uses AES-128 (Advanced Encryption Standard) link layer security, which is defined in IEEE 802.15.4 and it provides link authentication & encryption.

Basic Requirements of 6LoWPAN

- The device should be having sleep mode in order to support the battery saving.
- Minimal memory requirement.
- Routing overhead should be lowered.

Features of 6LoWPAN

- It is used with IEEE 802.15,.4 in the 2.4 GHz band.
- Outdoor range: ~200 m (maximum)
- Data rate: 200kbps (maximum)
- Maximum number of nodes: ~100

Advantages of 6LoWPAN

- 6LoWPAN is a mesh network that is robust, scalable, and can heal on its own.
- It delivers low-cost and secure communication in IoT devices.
- It uses IPv6 protocol and so it can be directly routed to cloud platforms.
- It offers one-to-many and many-to-one routing.
- In the network, leaf nodes can be in sleep mode for a longer duration of time.

Disadvantages of 6LoWPAN

- It is comparatively less secure than Zigbee.
- It has lesser immunity to interference than that Wi-Fi and Bluetooth.
- Without the mesh topology, it supports a short range.

Applications of 6LoWPAN

- It is a wireless sensor network.
- It is used in home-automation,
- It is used in smart agricultural techniques, and industrial monitoring.
- It is utilised to make IPv6 packet transmission on networks with constrained power and reliability resources possible.

Security and Interoperability with 6LoWPAN

- Security: 6LoWPAN security is ensured by the AES algorithm, which is a link layer security, and the transport layer security mechanisms are included as well.
- Interoperability: 6LoWPAN is able to operate with other wireless devices as well which makes it interoperable in a network.

ZigBee Technology Architecture and Its Applications

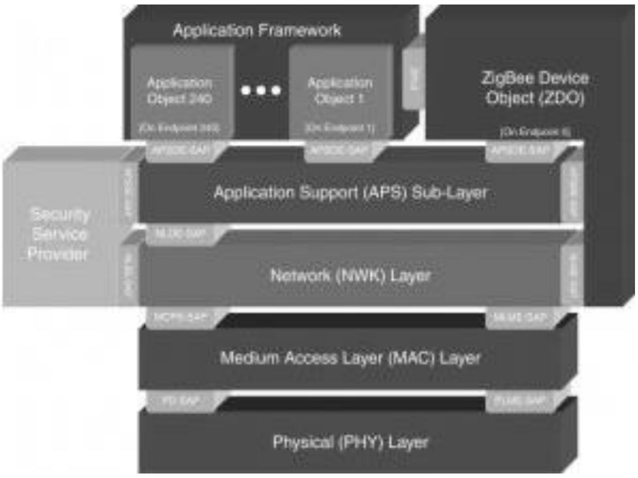
In this present communication world, there are numerous high data rate communication standards that are available, but none of these meet the sensors' and control devices' communication standards. These high-data-rate communication standards require low-latency and low-energy consumption even at lower bandwidths. The available proprietary wireless systems' Zigbee technology is low-cost and low-power consumption and its excellent and superb characteristics make this communication best suited for several embedded applications, industrial control, and home automation, and so on. The Zigbee technology range for transmission distances mainly ranges from 10 – 100 meters based on the output of power as well as environmental characteristics.

What is Zigbee Technology?

Zigbee communication is specially built for control and sensor networks on IEEE 802.15.4 standard for wireless personal area networks (WPANs), and it is the product from Zigbee alliance.
This communication standard defines physical and Media Access Control (MAC) layers to handle many devices at low-data rates. These Zigbee's WPANs operate at 868 MHz, 902-928MHz, and 2.4 GHz frequencies. The data rate of 250 kbps is best suited for periodic as well as intermediate two-way transmission of data between sensors and controllers.
Zigbee is a low-cost and low-powered mesh network widely deployed for controlling and monitoring applications where it covers 10-100 meters within the range. This communication system is less expensive and simpler than the other proprietary short-range wireless sensor networks as Bluetooth and Wi-Fi.

Zigbee Architecture

Zigbee system structure consists of three different types of devices as Zigbee Coordinator, Router, and End device. Every Zigbee network must consist of at least one coordinator which acts as a root and bridge of the network. The coordinator is responsible for handling and storing the information while performing receiving and transmitting data operations.
Zigbee routers act as intermediary devices that permit data to pass back and forth through them to other devices. End devices have limited functionality to communicate with the parent nodes such that the battery power is saved as shown in the figure. The number of routers, coordinators, and end devices depends on the type of networks such as star, tree, and mesh networks.
Zigbee protocol architecture consists of a stack of various layers where IEEE 802.15.4 is defined by physical and MAC layers while this protocol is completed by accumulating Zigbee's own network and application layers.



ZigBee Technology Architecture

Physical Layer: This layer does modulation and demodulation operations upon transmitting and receiving signals respectively. This layer's frequency, data rate, and a number of channels are given below.
MAC Layer: This layer is responsible for reliable transmission of data by accessing different networks with the carrier sense multiple access collision

avoidances (CSMA). This also transmits the beacon frames for synchronizing communication.
Network Layer: This layer takes care of all network-related operations such as network setup, end device connection, and disconnection to network, routing, device configurations, etc.
Application Support Sub-Layer: This layer enables the services necessary for Zigbee device objects and application objects to interface with the network layers for data managing services. This layer is responsible for matching two devices according to their services and needs.
Application Framework: It provides two types of data services as key-value pair and generic message services. The generic message is a developer-defined structure, whereas the key-value pair is used for getting attributes within the application objects. ZDO provides an interface between application objects and the APS layer in Zigbee devices. It is responsible for detecting, initiating, and binding other devices to the network.

Zigbee Operating Modes and Its Topologies

Zigbee two-way data is transferred in two modes: Non-beacon mode and Beacon mode. In a beacon mode, the coordinators and routers continuously monitor the active state of incoming data hence more power is consumed. In this mode, the routers and coordinators do not sleep because at any time any node can wake up and communicate.
However, it requires more power supply and its overall power consumption is low because most of the devices are in an inactive state for over long periods in the network. In a beacon mode, when there is no data communication from end devices, then the routers and coordinators enter into a sleep state. Periodically this coordinator wakes up and transmits the beacons to the routers in the network.
These beacon networks are work for time slots which means, they operate when the communication needed results in lower duty cycles and longer battery usage. These beacon and non-beacon modes of Zigbee can manage periodic (sensors data), intermittent (Light switches), and repetitive data types.

Zigbee Topologies

Zigbee supports several network topologies; however, the most commonly used configurations are star, mesh, and cluster tree topologies. Any topology consists of one or more coordinators. In a star topology, the network consists of one coordinator which is responsible for initiating and managing the devices over the network. All other devices are called end devices that directly communicate with the coordinator.
This is used in industries where all the endpoint devices are needed to communicate with the central controller, and this topology is simple and easy to deploy. In mesh and tree topologies, the Zigbee network is extended with several routers where the coordinator is responsible for staring them. These structures allow any device to communicate with any other adjacent node for providing redundancy to the data.
If any node fails, the information is routed automatically to other devices by these topologies. As redundancy is the main factor in industries, hence mesh topology is mostly used. In a cluster-tree network, each cluster consists of a coordinator with leaf nodes, and these coordinators are connected to the parent coordinator which initiates the entire network.
Due to the advantages of Zigbee technology like low cost and low power operating modes and its topologies, this short-range communication technology is best suited for several applications compared to other proprietary communications, such as Bluetooth, Wi-Fi, etc. some of these comparisons such as range of Zigbee, standards, etc., are given below.

The advantages of Zigbee include the following.

- This network has a flexible network structure
- Battery life is good.

- Power consumption is less
- Very simple to fix.
- It supports approximately 6500 nodes.
- Less cost.
- It is self-healing as well as more reliable.
- Network setting is very easy as well as simple.
- Loads are evenly distributed across the network because it doesn’t include a central controller
- Home appliances monitoring as well controlling is extremely simple using remote
- The network is scalable and it is easy to add/remote ZigBee end device to the network.

The disadvantages of Zigbee include the following.

- It needs the system information to control Zigbee based devices for the owner.
- As compared with WiFi, it is not secure.
- The high replacement cost once any issue happens within Zigbee based home appliances
- The transmission rate of the Zigbee is less
- It does not include several end devices.
- It is so highly risky to be used for official private information.
- It is not used as an outdoor wireless communication system because it has less coverage limit.
- Similar to other types of wireless systems, this ZigBee communication system is prone to bother from unauthorized people.

Applications of Zigbee Technology

The applications of ZigBee technology include the following.

Industrial Automation: In manufacturing and production industries, a communication link continually monitors various parameters and critical equipment. Hence Zigbee considerably reduces this communication cost as well as optimizes the control process for greater reliability.
Home Automation: Zigbee is perfectly suited for controlling home appliances remotely as a lighting system control, appliance control, heating, and cooling system control, safety equipment operations and control, surveillance, and so on.
Smart Metering: Zigbee remote operations in smart metering include energy consumption response, pricing support, security over power theft, etc.
Smart Grid monitoring: Zigbee operations in this smart grid involve remote temperature monitoring, fault locating, reactive power management, and so on.
ZigBee technology is used to build engineering projects like wireless fingerprint attendance system and home automation.